TOTAL SURVEY ERROR / TOTAL SURVEY QUALITY II

Class 1
Jan 13, 2026

Welcome back !

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My mom who is 63 is pursuing an associates degree. Today she asked me if it’s normal to forget what she studied after the semester ends.

Today’s session

1. Why 721?

2. Syllabus/Course overview

3. Scientific context

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Why 721?

To give you experience conducting **independent methodological research**

- Explore a methodological problem related to a type of survey error
- Conduct analysis that sheds light on that error source
  - Why does this error matter? Is this a big deal?
  - What causes it?
  - What can we do about it?
  - How effective are different solutions?
- **Writing a paper in a journal article format**
  - The paper should be a substantial piece of work (~thesis)
    - density of ideas more important than number of pages or words
  - Several TSE/TSQ papers have been published!
- Many intermediate assignments leading to the final paper
- Feedback from peers, research partner, and instructor

But I am not interested in ‘research’…

But I have no ideas…

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Getting the paper done

https://scientificwritingtips.wordpress.com/the-cartoons/

- Assignments 1-8, culminating in the final paper
- Instructor feedback on Assignments 1-8
- Research partner feedback on Assignment 2-8
- Assignments designed to keep you moving forward and to provide plenty of timely feedback
- If you keep up, you'll get ample feedback from the instructor and your partner and you'll finish the paper
- Review the syllabus.

Group Presentation

- Understanding paper structures
- How are published papers structured?
- What are the stories in the paper? What is the KEY story in the paper?
- What role does each section play in delivering the stories?
- **What gap in methodological knowledge / methodological weakness is being addressed in the paper?**

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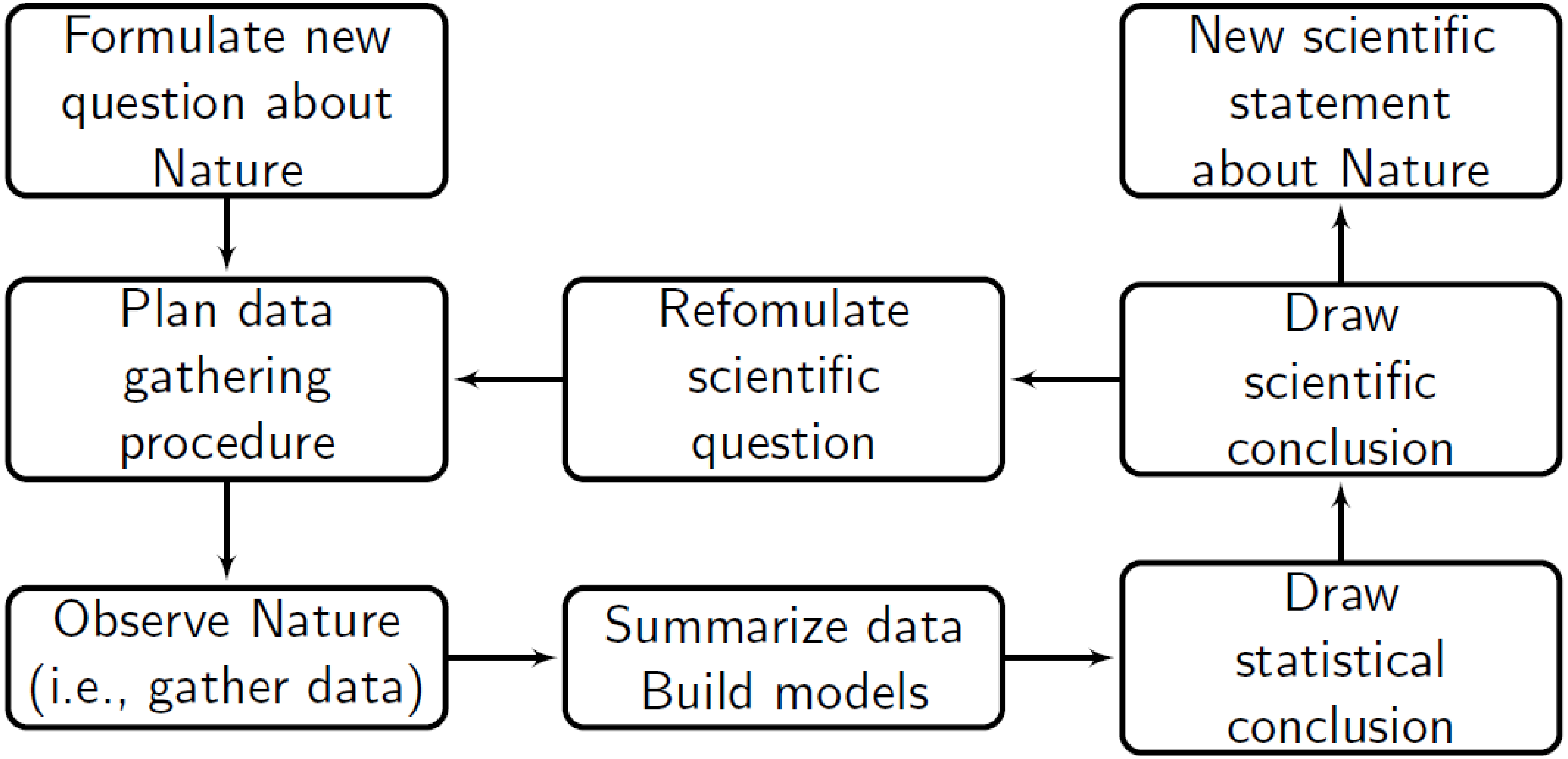
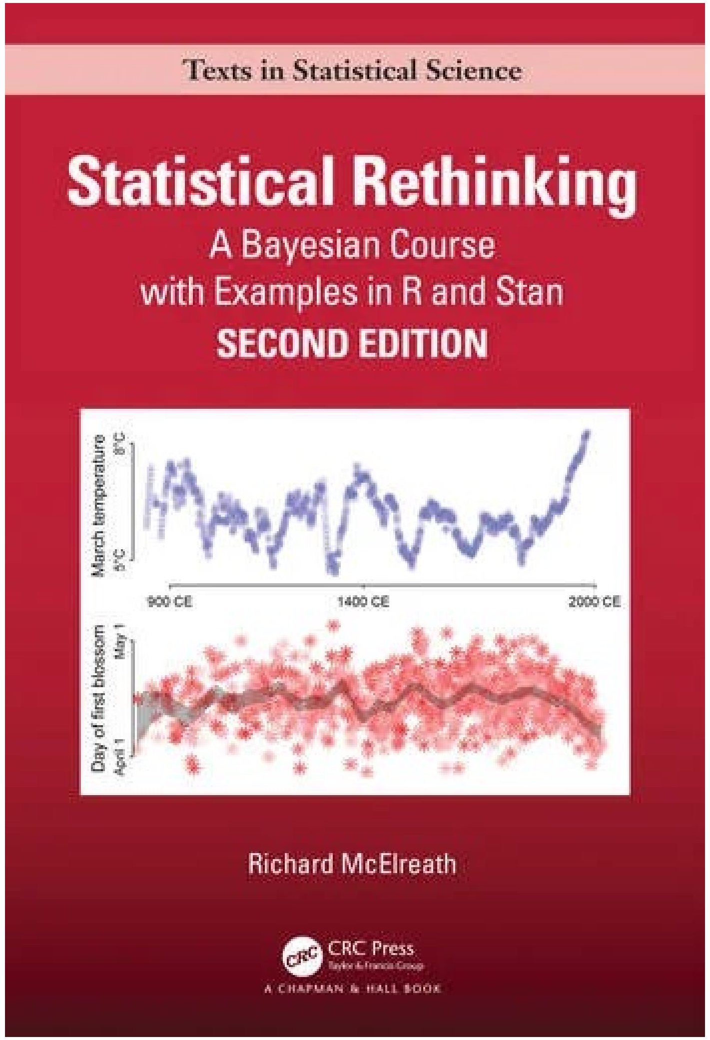
Scientific Reasoning



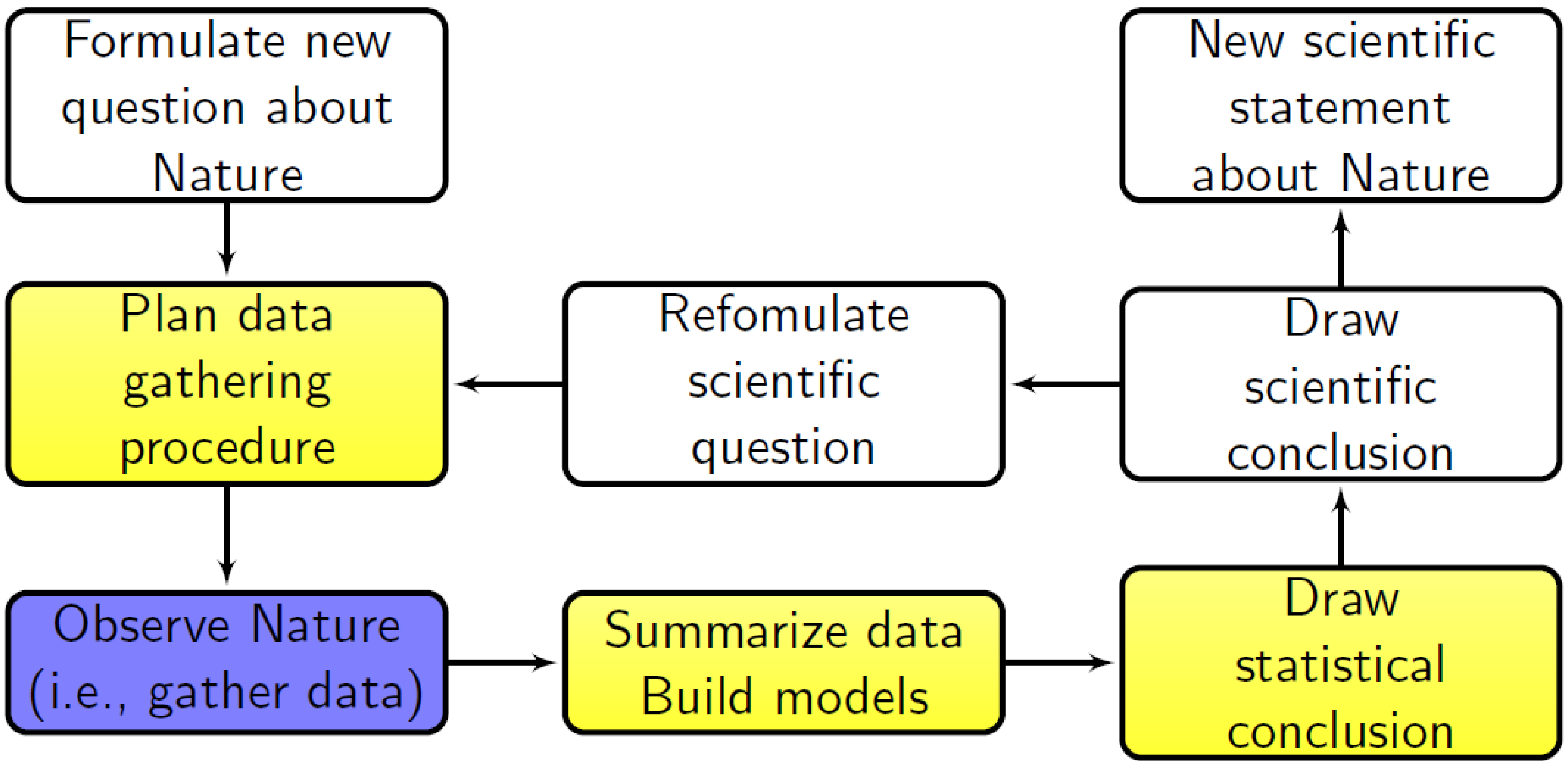


1.2.1. Hypotheses are not models. When we attempt to falsify a hypothesis, we must work with a model of some kind. Even when the attempt is not explicitly statistical, there is always a tacit model of measurement, of evidence, that operationalizes the hypothesis. All models are false,[6] so what does it mean to falsify a model? One consequence of the requirement to work with models is that it's no longer possible to deduce that a hypothesis is false, just because we reject a model derived from it.

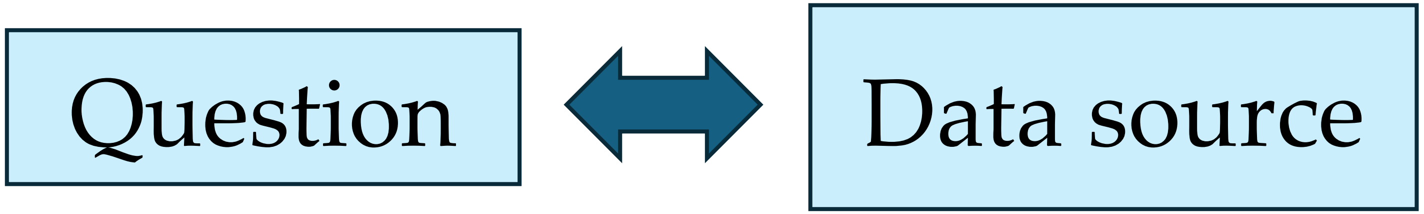
The SMALL WORLD is the self-contained logical world of the model. Within the small world, all possibilities are nominated. There are no pure surprises, like the existence of a huge continent between Europe and Asia. Within the small world of the model, it is important to be able to verify the model's logic, making sure that it performs as expected under favorable assumptions. Bayesian models have some advantages in this regard, as they have reasonable claims to optimality: No alternative model could make better use of the information in the data and support better decisions, assuming the small world is an accurate description of the real world.[39]

The LARGE WORLD is the broader context in which one deploys a model. In the large world, there may be events that were not imagined in the small world. Moreover, the model is always an incomplete representation of the large world, and so will make mistakes, even if all kinds of events have been properly nominated. The logical consistency of a model in the small world is no guarantee that it will be optimal in the large world. But it is certainly a warm comfort.

Survey Methodology in Scientific Reasoning



How does one come up with a research question?



Things to Consider

- Considerations when formulating a research question
  - Relevant
  - Substantial and original
  - Clear and simple
  - Manageable within available resources
  - Do you find the topic interesting?

- Approach
  - Familiarize yourself with relevant literature
  - Identify a gap in the research literature – something that is not known but that would be valuable for researchers to know

Different Types of Methodological Research

- Original data collection, often with methodological experiments
- Secondary data analysis, using “observational” methods to draw conclusions
- Statistical examination of the error in a particular survey (“applied research”)
- More theoretical examination of a statistical issue
- Research synthesis
- **We will consider examples!**

Different Types of Methodological Research

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Also Different Structures of Research Papers

- Technical reports
  - Documenting how a specific survey was done
  - e.g., Technical Report 63 on the CPS, "The CPS: Design and Methodology"

- Quality profile
  - Attempt to catalogue and, where possible, quantify different sources of error in a survey, e.g., BRFSS Summary Data Quality Report

- Scientific reports
  - Journal articles (short, our focus) or monographs (long)

Structure of Journal Articles

- Some variation across disciplines and by type of research being reported, but some common features as well
- Most journal articles follow the same general outline:
  - Abstract
  - Introduction
  - Methods
  - Results
  - Discussion
  - References
  - Appendices (if needed)
- Different professional organizations vary in conventions and formats for citing works in the text and reference list (e.g., AAPOR vs. ASA vs. APA)

Abstract

A brief (~200-300 words) summary of the contents of the article

- Some journals require a particular format, specifying what has to be covered, e.g., Background, Methods, Results, Conclusions
- Some journals require key words
- Most journals specify length (number of words)
- Try to make it comprehensible, jargon-free, citation-free

Re-Examining the Middle Means Typical and the Left and Top Means First Heuristics Using Eye-Tracking Methodology

Get access >

Jan Karem Höhne, Timo Lenzner, Cornelia E Neuert, Ting Yan

Journal of Survey Statistics and Methodology, Volume 9, Issue 1, February 2021, Pages 25–50, https://doi.org/10.1093/jssam/smz028

Published: 21 October 2019

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Abstract

Web surveys are a common self-administered mode of data collection using written language to convey information. This language is usually accompanied by visual design elements, such as numbers, symbols, and graphics. As shown by previous research, such elements of survey questions can affect response behavior because respondents sometimes use interpretive heuristics, such as the "*middle means typical*" and the "*left and top means first*" heuristics when answering survey questions. In this study, we adopted the designs and survey questions of two experiments reported in Tourangeau, Couper, and Conrad (2004). One experiment varied the position of nonsubstantive response options in relation to other substantive response options and the second experiment varied the order of the response options. We implemented both experiments in an eye-tracking study. By recording respondents' eye movements, we are able to observe how they read question stems and response options and we are able to draw conclusions about the survey response process the questions initiate. This enables us to investigate the mechanisms underlying the two interpretive heuristics and to test the assumptions of Tourangeau et al. (2004) about the ways in which interpretive heuristics influence survey responding. The eye-tracking data reveal mixed results for the two interpretive heuristics. For the *middle means typical* heuristic, it remains somewhat unclear whether respondents seize on the conceptual or visual midpoint of a response scale when answering survey questions. For the *left and top means first* heuristic, we found that violations of the heuristic increase response effort in terms of eye fixations. These results are discussed in the context of the findings of the original studies.

VOLUME 50 - ARTICLE 32 | PAGES 929–966

Gone and forgotten? Predictors of birth history omissions in India

BY Sharan Sharma, Sonalde Desai, Debasis Barik, Om Prakash Sharma

Abstract

Background: Fertility histories are subject to measurement errors such as incorrect birth dates, incorrect birth orders, incorrect sex, and omissions. These errors can bias demographic estimates such as fertility rates and child mortality rates.

Objective: We focus on births missing in fertility histories. We estimate the prevalence of such omissions and study their associated factors.

Methods: We leverage a panel survey (the India Human Development Survey) where the same women were interviewed in two waves several years apart. We compare data across waves and identify omitted births. Omissions in the second wave are modeled as a function of several child, mother, household, and survey interviewer variables. Models are fit separately to omissions reported alive or dead in the first wave.

Results: We conservatively estimate the prevalence of omissions at 4%. A large majority of omitted births are those of dead children, especially infants, with children in poorer households at greater risk of being omitted. For children alive in wave 1, female children are much more likely to be omitted in wave 2 compared to male children. Interviewers can detect respondent behaviors associated with omissions.

Conclusions: Omissions in fertility histories are non-ignorable. They do not randomly occur, and they affect some population subgroups and some interview contexts more than others.

Contribution: We investigate the understudied but important phenomenon of omitted births in fertility histories. We bring attention to possible biases in demographic estimates. We shed light on the survey process and propose strategies for minimizing bias through improved survey design.

Introduction

- This section provides the background for the study
  - Motivation — why does your study matter?
  - Literature review
    - What have previous investigators found?
    - Places your study in the context of earlier work
      - Why is this an important problem? (GAP / WEAKNESS in the literature)
      - What is the contribution of your study to the literature on this problem?

- Often includes some hypotheses that your study will test
  - Verbal, substantive assertion: "Incentives increase response propensities, with a larger impact on respondents with little interest in the topic of the survey"
  - Statistical statement: "In general, we expect incentives to have a main effect and to interact with topic interest"
  - Sometimes explicitly labelled: "H1: Incentives increase response propensities …"

Introduction…

- The "hypothesis" may also take the form of a research question: "This study will compare two forms of imputation in a survey of establishments and ask which one produces lower mean squared error for key estimates?"
- The hypothesis/research questions may serve as a bridge to the more technical material

  - "To test this hypothesis, we carried out an experiment that varied both the topic of the survey and the incentives offered to sample members."

Methods

- The Methods section describes *how* your research was carried out and *what* exactly your study has done
- Generally, the Methods section covers

  - Where did the data come from?

    - If it's a secondary analysis, this might take the form of a relatively brief description of the survey (say, the CPS or GSS – cite the relevant report)

    - If it's a new study, you need to describe the sampling, data collection (giving response rates), experimental design, weighting procedures, etc.

  - What were the most important procedures, e.g., how you did your simulations, how you dealt with missing data), this should be covered here

  - A detailed description of the analytic plan should also be included

  - Issues pertaining to the results of a specific analysis should be discussed in the Results; keep these distinct!

  - Wording of key survey items typically provided

Results

- This section describes what you learned from your analyses/simulations/experiments
- Shows analytical results in detail
- Sometimes organized by study (if there are multiple data sets)
- Sometimes organized by outcome measures or research questions
- Should follow good practices for tables and figures
- Should report standard errors beside point estimates and appropriate significance tests
- **Interpret results systematically and with focus; do not discuss in more detail than necessary – especially if not relevant to hypotheses or research questions**
- **Do not interpret the results very much. That’s for the…**

Discussion

- This section moves back from the technical descriptions of the analyses in the Results section to the broader questions with which the research began
  - May include a summary of the key results, and where these results fit in the existing literature
  - **Should include a statement of the conclusions you've drawn**
- Often includes a discussion of the limitations of the research that may limit the conclusions
- May also discuss additional studies that ought to be done

References

- This is not a bibliography of the materials you consulted
- It is a list (in the appropriate format) of the works you mentioned in the write-up
- You should cite major works on the topic, especially ones that motivate what you're doing
- For example, you should cite papers that propose or test the theory on which your hypothesis was based
- Give credit (for a given theory, idea, analytical procedure) where credit is due
- It is not customary to cite developers of standard analytic procedures (say, ANOVA)

Key : Make an Argument, TELL A STORY

- Difference between documentation and scientific article

  - Documentation designed to be comprehensive; tables in government reports allow readers to retrieve specific estimates

  - Scientific article designed to reach some (hopefully, general) conclusion with theoretical significance

  - Population inferences vs. Causal conclusions

- It's okay to leave out (or describe very briefly) analyses that aren't relevant to the conclusions
- Similarly, the literature you cite needs to be related to the point that you are making

Finding Literature

- Sources: Search engines, Google Scholar, PubMed Central (PMC), Web of Science (Science Citation Index)
  - Avoid AI here
- Be creative with key words; disciplines vary in their use of terms (e.g., systematic error – bias; random error/noise – variance)
- Reverse citation search is a good start (who is citing the key paper)

Next Week…

- Present one of your research ideas in Assignment #1 to the class
  - 1 idea only
  - What is the question?
  - Why is this a big deal?
  - How would you design the study?
  - What (kind of) data will you use?
- 1 slide
- Four-minute presentation + Two minutes of class feedback

Questions?

- You should think of this class as an opportunity to gain hands-on experience producing (writing) an academic paper in survey methodology with very structured guidance!
- You should consider submitting your papers to conferences / journals / student competitions / etc. when you are done!
- Have fun !